Sustainable Materials and Technologies 44 (2025) e01335

Contents lists available at ScienceDirect

Sustainable Materials and Technologies

journal homepage: www.elsevier.com/locate/susmat

3D-printable PVA-based inks filled with leather particle scraps for UV-assisted direct ink writing: Characterization and printability

Luca Guida [a,*,1], Alessia Romani [a,b,1], Davide Negri [a], Marco Cavallaro [a], Marinella Levi [a]

[a] Department of Chemistry, Materials and Chemical Engineering "Giulio Natta", Politecnico di Milano, Piazza Leonardo Da Vinci 32, 20133 Milano, Italy.
[b] Department of Electrical and Computer Engineering, Faculty of Engineering, Western University, London, ON, Canada

ARTICLE INFO

Keywords:
Material extrusion (MEX)
Recycling
Polymer composites
Mechanical properties
Bio-based
Additive manufacturing

ABSTRACT

Despite its significant environmental impacts, leather remains a popular material due to its durability, aesthetics, and mechanical properties. Recycling leather scraps is gaining increasing attention to reduce waste, pollutants, and emissions from pristine raw materials in the tanning industry. Material Extrusion additive manufacturing represents a promising way to recycle leather byproducts as secondary raw materials for new applications. This paper investigates the characterization and printability of photo- and thermal-curable PVA-based inks for UV-assisted Direct Ink Writing filled with leather filler scraps from the tanning industry, i.e., leather shavings. As a cold extrusion process, Direct Ink Writing reduces energy consumption and maximizes the waste percentage content in new material formulations. The morphology and thermal properties of the leather filler were assessed before adding it to a novel cross-linkable PVA-based matrix, preparing inks containing up to 20 % wt. of leather scraps, leading to almost 40 % wt. after post-curing. Rheological tests showed a shear-thinning behavior of the formulations and a clear transition from solid-like to fluid-like behavior, followed by a quick recovery of the solid-like behavior, ensuring good printability, shape retention, and fidelity. UV crosslinking and post-curing led to robust polymer networks, reaching crosslinking degrees of ~90 %. According to the mechanical tests, the PVA-leather scrap inks exhibited mechanical properties broadly similar to virgin leather materials, i.e., 170 MPa elastic modulus, 13 % elongation at break, and 6 MPa stress at break. Scanning Electron Microscopy revealed a preferential alignment of the leather filler along the extrusion direction, confirming the reinforcing effect of the scrap particles. These results demonstrate the suitability of these inks as alternatives for new tailored applications in the leather industry, reducing virgin material usage through additive manufacturing.

1. Introduction

Leather is extensively utilized across various industries, particularly in the automotive and fashion sectors, representing one of the most traded commodities at the international level [1]. Despite advancements in synthetic alternatives, e.g., polyurethane (PU) [2,3], leather remains highly favored for its durability, aesthetics, and performances, e.g., unique mechanical properties [4]. Several factors influence its microstructure, typically connected to the specific animal, environmental and weather conditions, production processes, and parameter setups [4–6], modifying its mechanical behavior for several tailored applications. In 2022, the leather market was valued at USD 440.64 billion, showing an estimated growth of USD 738.61 billion within the next six years [7]. Nevertheless, leather production leads to significant resource consumption and environmental impacts. For instance, 15,000 m^3 to 120,000 m^3 are required to process 1 Mg of raw material, where only one-fourth becomes leather [8,9]. Its production generates pollutant emissions, such as heavy metals, acids, greenhouse gases, and volatile organic compounds, depending on the treatment procedures and technologies [10]. As an example, tanning, its main processing, can raise health and environmental concerns. It typically involves chromium oxidation, representing a cancer risk factor [11] and a possible groundwater and soil contaminator [12]. Moreover, leather manufacturing generates non-recyclable industrial byproducts, including shavings, dust, and fleshing waste, i.e., around 400–700 Kg of solid waste for producing 0.25 Mg of raw material [9,11].

These challenges have led the leather industry to focus on the implementation of new procedures and actions towards cleaner

* Corresponding author.
E-mail address: luca.guida@polimi.it (L. Guida).
[1] These authors equally contributed to this work.

https://doi.org/10.1016/j.susmat.2025.e01335
Received 6 December 2024; Received in revised form 17 February 2025; Accepted 26 February 2025
Available online 28 February 2025
2214-9937/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

production, including reducing water consumption and hazardous chemicals usage or improving wastewater treatment and solid-waste recovery [11,13]. In this context, the circular economy represents a key element in reducing the environmental impact of the leather industry value chain [9], helping to handle either big-sized scraps or small particles, e.g., shavings. Tanned and untanned leather scraps and byproducts are currently considered as resources to be valorized as secondary raw materials, supporting emission control and cost-saving for waste management and reducing the use of virgin materials [10]. For instance, they can be reused in the cosmetic and pharmaceutical industries for their protein, fat, and water content [14], or as a source for biofuel production [15], as well as for organic fertilizers and additives in the agricultural sector. [16]. However, scraps from leather tanned with chromium or other synthetic agents, e.g., glutaraldehyde (GTA), cannot be directly composted and used for food additives or fertilizers due to the presence of residual chromium and toxic chemical species [17]. In addition, the hydrolysis process necessary for their use in such sectors often produces low-purity products or is relatively slow and costly. For this reason, alternative circular paths should be detected to retain the value of these scraps [17].

Another possible application is their use for the development of new materials [18,19]. Leather scraps can be reused in multiple ways, requiring different reprocessing steps depending on the size and quality of the byproducts, e.g., trimmings from leather sheets or shaving particles. From the literature, these secondary raw materials can be repurposed for different applications, involving a wide range of processes and steps according to the type and shape of the recycled leather feedstock. For instance, leather trimming waste has been used to produce sound-absorbing multilayer materials, using electrospinning to fabricate nanofiber layers starting from the collagen hydrolysate extracted from leather byproducts [20]. Surfactant agents have been produced from leather protein fraction residues through acid hydrolysis [21]. Other studies focused on reusing leather scraps and shavings with minimal reprocessing of the starting byproduct, fitting with the circular economy principles of resource efficiency and waste reduction [22,23]. Fewer reprocessing steps can help (i) preserve material value and quality, retaining its starting properties; (ii) limit energy, water, and resource use during byproduct transformation, as well as emissions; and (iii) foster cost-effective and local reuse, increasing the economic viability. To this end, leather scraps can be used either as matrixes or as reinforcements for new composite materials, especially in polymer-based composites [24–26]. Previous work successfully used leather waste with different thermoplastics, such as Acrylonitrile Butadiene Styrene (ABS) [27,28], Polyurethane (PU) [29], High-Density Polyethylene (HDPE) [30–32], and Polyethylene-vinyl acetate (EVA) [33,34]. These studies demonstrate the versatility of leather byproducts in enhancing different properties, e.g., mechanical performances, flame retardant, and heat insulation behaviors [32,35,36], together with their potential role in virgin resource retention.

Several conventional manufacturing processes can be used to fabricate new parts and products with these materials, including compression and injection molding [19,27,31]. Despite their potential, current solutions mostly combine leather scraps with petroleum-based thermoplastics, with only a few examples matching them with bio-based matrixes. Furthermore, the scrap percentages in the material formulations are still limited, i.e., usually between 10 and 30 % in weight. [27,32], and the melt compounding with thermoplastic polymers can have a significant energetic impact [37]. Additive Manufacturing (AM) represents a valid option for such composite materials to enlarge the range of potential sectors thanks to its capability to produce small batches of customized freeform parts [38–40]. Some AM technologies, in particular Material Extrusion (MEX) processes [41], have been exploited with different types of scraps and byproducts, ranging from recycled thermoplastics [42,43], agro-industrial biomass [44], and fiber-reinforced plastics [45,46]. Although Fused Filament Fabrication (FFF) is one of the most popular AM processes when dealing with secondary raw materials, Direct Ink Writing (DIW) further expands the range of recycled feedstocks for new applications with MEX AM. This process allows the extrusion of viscous pastes, hydrogels, inks, or cross-linkable resins at room temperature through a nozzle or orifice, passing from a liquid-like behavior when applying a shear force to a solid-like after its deposition [47], enabling an energetically-efficient process. DIW allows the reprocessing of several virgin and recycled feedstocks otherwise not available for MEX AM, including biomass-based pastes [48], ceramic scraps [49], and fiber-reinforced composites [50]. Thanks to its cold-extrusion nature, DIW can control particle alignment during fabrication [51,52] and help preserve the original properties of the selected scrap, avoiding thermomechanical degradation due to hotend feedstock melting [44].

Furthermore, DIW enables the processing of monomeric resins or solvent-based polymer solutions, which can incorporate fillers of various types. In the case of solvent-based solutions, this technique facilitates the production of dry materials with a higher filler content after evaporation, as the solvent initially aids in extrusion but subsequently evaporates [53]. Indeed, the filler content in the final product has been reported to exceed 30 % for water-based solutions [54,55], whereas it is typically lower when using monomeric resins [56–58].

Leather-like materials and inks can also be manufactured by MEX AM and DIW, leveraging the use of alternative raw materials, e.g., silk proteins [59,60]. Despite the increasing interest in AM for recycled biomass feedstocks [44,61], abundant leather scraps and byproducts have not been currently considered as a potential secondary raw material for MEX AM and DIW. Nevertheless, these processes represent a possible way to close the loop of the tanning industry and reduce the use of virgin resources, maximizing the scrap content percentage and allowing the fine-tuning of the final part properties. According to the available literature, no works currently explore the potential application of leather waste for DIW and, in general, other AM technologies.

This study explores the use of AM as a tool for recycling secondary raw materials derived from well-established industrial supply chains, focusing on waste from the tanning industry. The research develops and characterizes polymer-filler composites incorporating leather waste particles (LF) from industrial processes. Poly(vinyl alcohol) (PVA)-based materials, designed for photo- and thermal-curing, were formulated as inks for UV-assisted DIW. The selected technology is particularly suitable due to its potential to maximize scrap percentage content and reduce energy consumption, being a room-temperature extrusion process [62], while PVA matrices have already been used in literature as an environmentally friendly polymer [63,64]. Their printability and performance were systematically evaluated. At a later stage, the goal is to obtain a material formulation comparable with virgin leather in terms of mechanical properties, thereby reducing reliance on environmentally impactful raw materials and expanding the application potential of DIW technology to include recycled leather. Using a water-based PVA solution for the process allowed to maximize the use of recycled materials and reduce the need for virgin resources, achieving a scraps dry fraction as high as 38 % *w*/w with no particular issues related to the processability of the compositions with DIW technology.

After assessing the morphology and thermal properties of the LF (Subsection 3.1), a novel photo- and thermal-curable PVA-based formulation was synthesized as a matrix to be used with the LF for the UV-assisted DIW process (Subsection 3.2). The rheological properties of three different PVA-leather inks were evaluated to understand their suitability for the DIW process (Subsection 3.3). The formulations were then 3D printed to evaluate their printability and measure the shape fidelity and retention of the extruded paths (Subsection 3.4). Differential Scanning Calorimetry (DSC) and gel content measurements were then carried out to evaluate the curing process and the crosslinking degree of the formulations for UV-assisted DIW (Subsection 3.5). At the end, tensile tests were conducted on cast and 3D printed specimens to assess the mechanical properties for potential applications, analyzing the morphology of the fracture surfaces to give further insight into the reinforcing nature of the LF (Subsection 3.6). Given the cold-extrusion nature of the 3D printing process, the effect of the selected processing technology has been assessed based on morphological considerations through analysis of sample cross-sections [65,66] and mechanical properties [67]. Indeed, the only difference between cast and printed samples was expected to be due to the alignment of particles during the extrusion [51,52]. According to the results, PVA-LF inks can be processed through UV-assisted DIW, achieving good printability, shape fidelity, and crosslinking degrees. A more extensive experimental campaign could be conducted to identify optimal formulations for the intended purpose, and alternative matrices could be explored to achieve, for instance, flexible behavior. Nonetheless, the formulations developed exhibited mechanical properties comparable to those of virgin leather, making them a promising alternative for novel applications to reduce reliance on pristine raw materials.

2. Materials and methods

2.1. Materials

Milled Glutaraldehyde-tanned Leather Shavings with a maximum granulometry of 4 mm were provided by the Italian Leather Research Institute (SSIP, Pozzuoli, Naples, Italy). The filler was sieved through a 0.5 mm sieve before use to increase granulometry homogeneity, favor proper extrudability, and limit nozzle clogging.

Polyvinyl alcohol (PVA, M_w 31,000–50,000, 98–99 % hydrolyzed), glycidyl methacrylate (GMA), *N*,*N*,*N*′,*N*′-Tetramethylethylenediamine (TEMED), deuterated dimethyl sulfoxide, glycerol, 2-Hydroxy-4′-(2-hydroxyethoxy)-2-methylpropiophenone (Irgacure2959, I2959), dicumyl peroxide and silica nano-powder (fumed, 200–300 nm) were purchased from Sigma Aldrich (Inc. St. Louis, MO. US). Laboratory-grade dimethyl sulfoxide (DMSO) and acetone were used as solvents. All chemicals were used as received without any further purification.

2.2. Characterization workflow and nomenclature

The characterization workflow followed in this work is shown in Fig. 1. The PVA matrix, here defined as PVA-M, underwent a functionalization step of methacrylation to make it suitable for cross-linking. PVA and PVA-M were analyzed through Nuclear Magnetic Resonance (NMR, Subsection 2.4). LF was analyzed in terms of morphology and then sieved to achieve a homogeneous particle granulometry (Subsection 2.3), hereinafter called LF-S, and used for all the formulations. LF-S was then analyzed in terms of morphology and thermal properties by means of DSC and Thermogravimetric Analysis (TGA).

As explained in Subsection 2.5, four different PVA-LF inks were tested after mixing PVA-M and LF-S with water, glycerol, the photo-initiator (I2959), and the thermal-initiator (dicumyl peroxide). For the sake of simplicity, each formulation was renamed "PVAX", where X refers to the w/w_{matrix} ratio of LF-S added to the specific ink, i.e., PVA20, PVA25, and PVA30. A benchmark formulation was also prepared, PVA0S, by adding 15 % w/w_{matrix} of silica nano-powder instead of LF-S. The four formulations underwent rheological tests (Subsection 2.6) to assess their suitability for UV-assisted DIW, and they were then 3D printed to evaluate their printability (Subsection 2.7) and the curing process through DSC and gel contents (Subsection 2.8). Tensile samples were produced either by casting or UV-assisted DIW to understand the influence of LF-S and the 3D printing process on the mechanical properties (Subsection 2.9). At this stage, each batch of samples was renamed "PVAXY", where X refers to the w/w_{matrix} ratio of LF-S added to the specific ink, and Y corresponds to the manufacturing process used to produce the specific batch of samples, i.e., "C" for casting and "P" for 3D printing. The morphology of the most promising formulation, PVA25, was then assessed from the fracture surfaces of the cast and 3D printed tensile specimens through Scanning Electron Microscopy (SEM) images (Subsection 2.9). A 3D printed square sample was finally 3D printed as a proof-of-concept with the same formulation (Subsection 2.7).

2.3. Leather filler

Leather filler was morphologically characterized both before and after sieving. LF was dispersed in ethanol by vigorous agitation with a 1:1000 *w/v* ratio. The resulting suspension was then placed onto a glass slide and allowed to settle, followed by drying at room temperature. Isolated particles from the dispersion were subsequently observed using

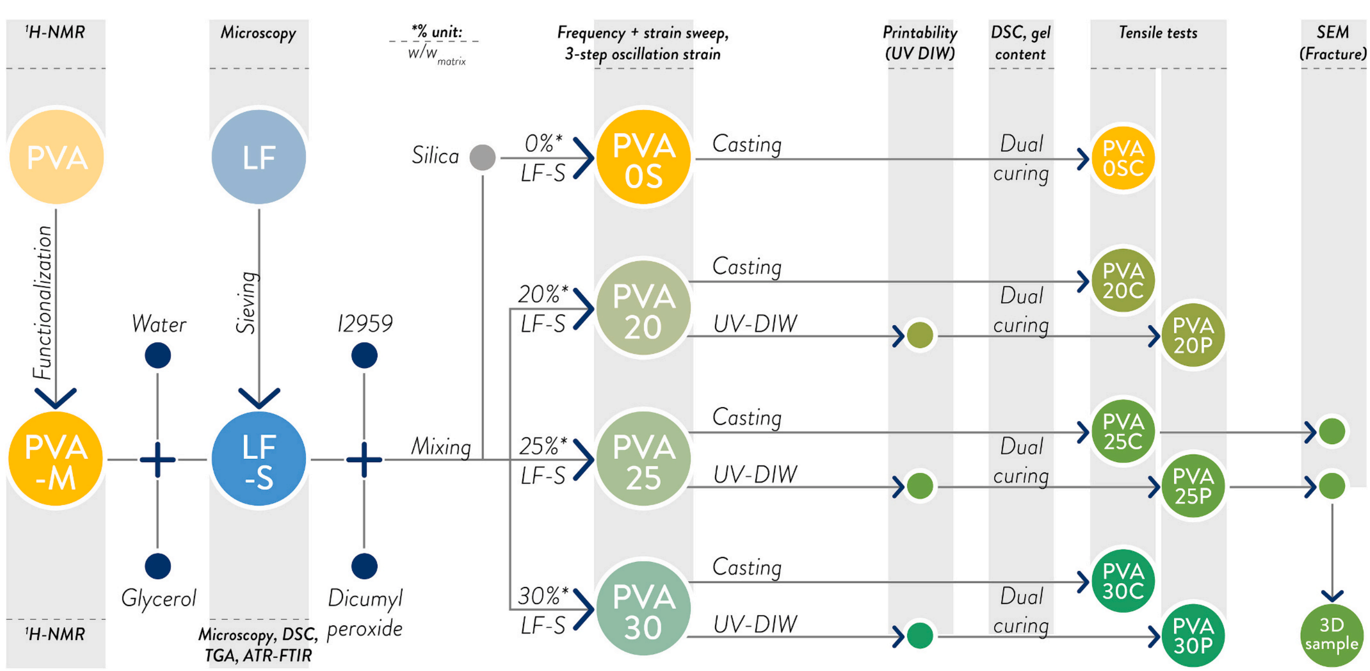

Fig. 1. Flow chart of the characterization conducted in this work with the different nomenclature of each sample batch (PVA-M: Methacrylated PVA; LF-S: Sieved Leather Filler). The percentage values of the LF-S contents are w/w_{matrix}.

an optical microscope at 50× magnification. (Olympus BX60 optical microscope, Olympus, Shinjuku-ku, Tokyo, Japan). A total of 60 images were captured for both the sieved and non-sieved LF samples. Images were analyzed using ImageJ open source software [68].

Given the irregular shape of the particles, their characteristic dimension was defined as the diameter of the circumscribed circle around each particle. A shape factor (SF), equal to the ratio of the particle area to the area of the circumscribed circle, was defined to describe the geometry of the particles. SF tends to be 1 for perfectly round particles and 0 for fibers and rod-like ones. The distribution of diameters and shape factors was finally fit using a Weibull distribution function, with a probability density function PD defined as follows (Eq. 1):

$$PD(x) = \begin{cases} \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-\frac{x^k}{\lambda}} & if\ x > 0 \\ 0 & if\ x < 0 \end{cases} \tag{1}$$

Where λ is the scale factor and k is the shape factor. This model is often used to describe the dimension of sieved particles since it can describe asymmetric and skewed distributions [69].

The thermal properties of sieved LF were evaluated through DSC and TGA. DSC analyses were performed with a DSC 823e Mettler-Toledo instrument (Mettler Toledo, Columbus, OH, US) under a nitrogen atmosphere. The heating ramp was set from 20 °C to 250 °C with a 20°/min heating rate. TGA was performed with a TA Instruments Q500 machine (TA Instruments Inc., New Castle, DE, US) under a nitrogen atmosphere. The samples (~20 mg) were subjected to a thermal cycle from 25 to 800 °C at a heating rate of 20 °C/min to assess the thermal stability of the leather shavings.

The chemical composition of sieved LF was investigated through Attenuated Total Reflection Fourier Transform Infrared Spectroscopy (ATR-FTIR) with a Nicolet iS50 FTIR Spectrometer (Thermo fisher Scientific, MA, US) equipped with a diamond-based window. Samples were dried in a vacuum oven at 60C for 12 h prior to every characterization.

2.4. Matrix synthesis

PVA was methacrylated according to a protocol described in the literature [70,71]. Briefly, PVA was dissolved in dimethyl sulfoxide with a concentration of 10 % *w/v* through moderate stirring at 60 °C. After complete dissolution, TEMED (added as a catalyst) and GMA (grafting reagent) were added dropwise with a TEMED/-OH molar ratio equal to 1 % and GMA/-OH molar ratio equal to 5 %. The reaction was carried out for 24 h at 60 °C under continuous stirring (Fig. 2).

The solution was then cooled to room temperature and added dropwise into an excess of acetone, maintaining an acetone/DMSO ratio of 10:1. When introduced to acetone, methacrylated PVA (PVA-M) precipitated as flakes, while the reaction residues dissolved in the acetone. The PVA-M flakes were then collected and washed with acetone three times to ensure thorough removal of the reaction residues. PVA-M was finally dried in a vacuum oven (60 °C, overnight) and stored at room temperature until use.

The methacrylation degree of PVA-M, expressed as the fraction of -OH groups substituted by methacrylic pendants, was characterized by proton NMR (^{1}H NMR) at 400 MHz with a Bruker AV 400 SampleXpress machine (Bruker Corporation, Billerica, MA, US). Samples were prepared by dissolving 5 mg of polymer in 600 μL deuterated DMSO. Spectra were processed using the software MestreNova NMR (Mestrelab Research S.L., Santiago de Compostela, Spain).

2.5. Composite formulation

The matrix for the composite formulations was prepared by dissolving PVA-M in water at a ratio of 1:3 *w/v* by stirring at 80 °C. LF-S (20 %, 25 %, and 30 % w/w_{matrix}) and glycerol (plasticizer, 20 % w/w_{matrix}) were then added. Finally, I2959 (photo-initiator, 3 % w/w_{matrix}) and dicumyl peroxide (thermo-initiator, 0.3 % w/w_{matrix}) were dissolved into the mixture. It is noteworthy that, after sample drying, the content of the filler becomes higher than the nominal one, reaching almost 40 % w/w_{tot}.

One formulation containing only silica nano-powder (15 % w/w_{matrix}) as a filler was prepared as a control material.

Table 1 Compositions of all the formulations prepared with PVA-M and LF-S and tested in this work.

2.6. Rheological behavior characterization

Rheological tests have been performed on the formulations to assess their rheological behavior and evaluate their extrudability. All tests were carried out with Discovery HR-2 rotational rheometer (TA Instruments Inc., New Castle, DE, USA) with a 20 mm plate-plate geometry at 25 °C with a 0.8 mm gap. Three different tests were conducted on the different formulations: a frequency sweep test, a strain sweep test, and a 3-step oscillation strain test.

A frequency sweep test was conducted to assess shear-thinning behavior, as high sample viscosity prevented proper flow tests. Samples were subjected to 1 % oscillation strain with frequency ranging logarithmically from 0.1 Hz to 100 Hz, recording 3 points per decade.

A strain sweep test was used to evaluate the yield point by applying an oscillatory strain logarithmically from 0.1 % to 100 % (oscillation frequency 1 Hz, recording 10 points per decade). The yield point was defined as the intersection of the loss and storage modulus.

To simulate the extrusion process and assess the return to a solid-like

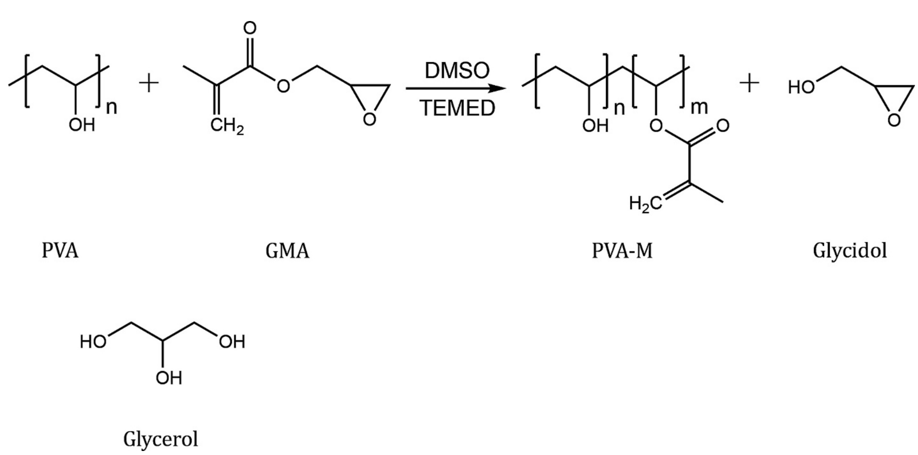

Fig. 2. PVA methacrylation reaction scheme.

Table 1
reports the compositions of all formulations prepared.

Formulation	Water [$w/w_{Tot,wet}$]	PVA-M [$w/w_{Tot,wet}$]	Silica [$w/w_{Tot,wet}$]	LF-S [$w/w_{Tot,wet}$]	LF-S [$w/w_{Tot,Dry}$]	Glycerol [$w/w_{Tot,wet}$]	I2959 [$w/w_{Tot,wet}$]	Dicumyl peroxide [$w/w_{Tot,wet}$]
PVA20	52.34 %	17.45 %	–	13.96 %	29.28 %	13.96 %	2.09 %	0.21 %
PVA25	50.57 %	16.86 %	–	16.86 %	34.11 %	13.49 %	2.02 %	0.20 %
PVA30	48.92 %	16.31 %	–	19.57 %	38.31 %	13.05 %	1.96 %	0.20 %
PVA0S	54.23 %	18.08 %	10.85 %	–	–	14.46 %	2.17 %	0.22 %

state, a 3-step oscillation strain test was performed at 1 Hz: 1 % strain for 300 s (reservoir simulation, quasi-static situation), 100 % strain for 300 s (extrusion simulation, large deformation), and 1 % strain for 300 s (post-deposition) [72].

2.7. 3D printing and curing

All inks were 3D printed using a custom DIW platform, equipped with various extruders, capable of both pneumatic and mechanical extrusion, and with a UV-curing module. Such a system is an in-house designed and implemented cartesian machine based on marlin firmware, allowing for a maximum printing volume equal to 200x200x200 mm.

The UV curing module comprises an array of 10 LEDs emitting at 365 nm (Intelligent LED Solutions, Thatcham, United Kingdom; 180 mW power output, 55° radiance angle). This array has been optimized to produce a uniform light intensity of 25 mW/cm^2 over a 90 mm diameter circular surface.

PVA-leather composites were 3D printed using pneumatic extrusion, which facilitates handling highly viscous materials. The G-code files were generated using Prusa Slicer 2.5.0 (Prusa Research, Prague, Czech Republic) and then modified with custom scripts to adapt them to the specific hardware of the custom printer. These modifications include, for instance, adding specific routines after each layer. The main 3D printing parameters are resumed in Table 2. The 3D printing protocol involved a UV-curing step after the deposition of each layer. After 3D printing each layer, the build platform was positioned under the UV-curing module and exposed to UV light for 60 s. This process partially crosslinked the ink layer-by-layer, thereby maintaining construct stability and shape retention throughout the 3D printing process.

Samples for tensile tests were 3D printed for each formulation alongside two structures designed to evaluate the ink shape fidelity. [73] A 14G (D = 1.6 mm) disposable plastic tapered nozzle has been used for the process (Techcon Systems Inc., Cypress, CA, USA). The material was deposited with a 3D printing speed equal to 4 mm/s, and the layer height was set at 0.8 mm.

Samples for tensile tests were produced by depositing five layers, each following a toolpath aligned with the loading direction of the sample, as depicted in Fig. 3a. This process resulted in samples with dimensions of 10 × 8 × 4 mm. After 3D printing, the samples were air-dried for 24 h at 25 °C and then thermally cured at 140 °C for 3 h.

Table 2
3D printing parameters used for the different sample batches with the PVA-leather composite inks.

Sample batches	Parameters	Units	Values
Common parameters for all batches	Layer height	mm	0.8
	Flow	%	100
	Speed	mm/s	4
	Nozzle Diameter	mm	1.6
Tensile samples and square samples only	Infill	%	100
Tensile samples only	Perimeters	–	0
Square samples only	Perimeters	–	1
3 × 3 grids and filament fusion geometry only	Infill	%	0
	Perimeters	–	1

Regarding shape fidelity, two monolayer structures were 3D printed to assess the printability of the ink. The first structure is a 3 × 3 grid (Fig. 3b) [73] designed to evaluate the ink's ability to preserve internal pore-like features. The printability index, Pr, can be defined from the analysis of the 3D printed grid as follows:

$$Pr = \frac{p^2}{16\,A} \tag{2}$$

where p represents the perimeter of a pore, and A is its internal surface area. Pr is expressed as the average of the values corresponding to the nine pores in the grid.

The last structure, the filament fusion geometry reported in Fig. 3c, allows to quantify the capability of the ink to preserve adjacent filaments deposited at limited distances. The evaluation of the result considers the ratio f_s/f_t as a function of fd (Fig. 3c, right). The parameter f_d represents the distance between two adjacent filaments, f_t denotes their width in the rectilinear region, and f^s indicates their width in the curved region. Shape fidelity measurements were carried out by analyzing images of the samples using ImageJ open-source software.

A fourth geometry was then 3D printed as a proof-of-concept of multilayered structures achievable with the best PVA-leather composite formulation. To this end, a 10 × 10 mm square made of 5 layers was fabricated with the PVA25 formulation, allowing for a qualitative evaluation of the process.

2.8. Curing characterization

The curing extent obtained with the UV-assisted DIW process reported in Subsection 2.7 has been evaluated by measuring the gel content of the cured samples and by DSC.

ASTM standard test method D2765–16 (2016) was followed to evaluate the gel content of the samples. Test method C was followed. Samples were immersed in water for 24 h on a magnetic stirrer (with a proportion of 300 mL water for 2 g of sample). Subsequently, samples were dried in a vacuum oven at 70 °C. Samples were weighted every 6 h. They were considered dry when the weight was stable for two subsequent measurements.

The gel content measurement was performed both on LF alone and composite formulations. For LF, the gel content was evaluated as the ratio of the final weight to the original one. As regards the composites, the gel content GC has been evaluated as follows:

$$GC = \frac{w_f}{k \cdot w_0} \tag{3}$$

In which w_0 and w_f are the initial and final weight of the sample, respectively, while the correcting factor k accounts for the loss of glycerol and possible soluble fractions of LF during the immersion in water. K has been defined as:

$$k = (1 - \%Gly - (1 - GC_{LF}) \cdot \%LF) \tag{4}$$

With %Gly and %LF being the dry content of glycerol, whereas LF and GC_{LF} being the gel content of LF previously measured.

As regards DSC, the crosslinking yield has been evaluated by measuring the crosslinking enthalpy of pristine and cured samples. The measurement has been carried out while the temperature has been increased between 25 and 250 °C at a rate of 10 °C/min. The

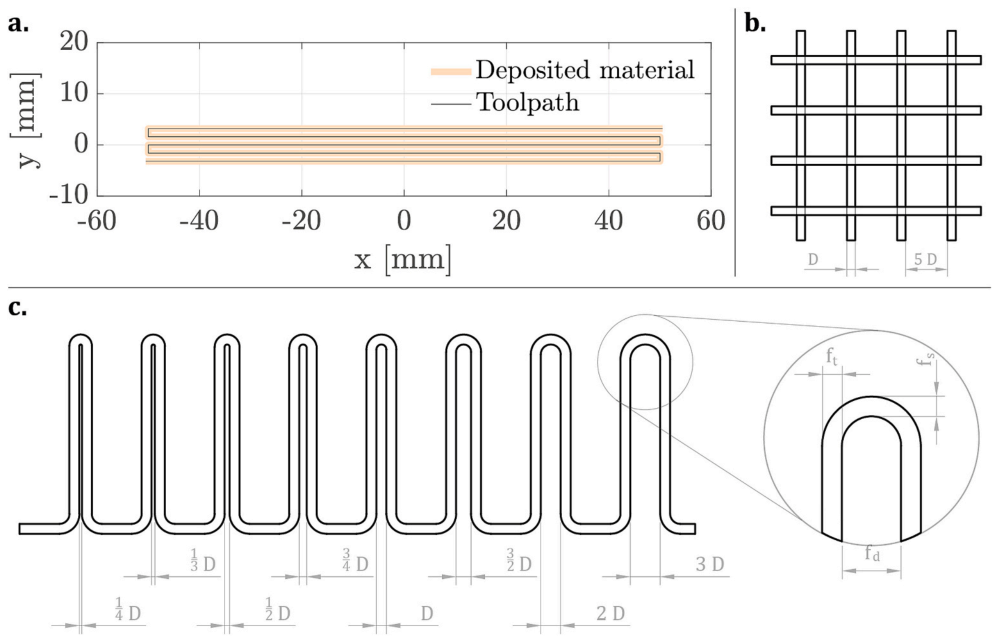

Fig. 3. Toolpaths of the 3D printed sample batches, D indicates the diameter of the nozzle: a) Toolpath followed to produce the tensile test samples; b) Geometry of the 3 × 3 grid for the evaluation of shape fidelity and printability index; c) Geometry for filament fusion test.

crosslinking degree has been evaluated as follows:

$$C\% = 1 - \frac{\Delta H_{PC}}{\Delta H \cdot \%PVA} \tag{5}$$

Where ΔH_{PC} is the residual enthalpy of the cured sample, and ΔH is the crosslinking enthalpy of the uncured sample.

2.9. Mechanical characterization

The tensile test was conducted on the cured samples following the ASTM standard test method D3039/D3039M–17 (2017) [74]. The nominal gauge length was set to 40 mm, with specimens having an overall length of 100 mm, a width of 8 mm, and a thickness of 4 mm. Samples were produced both via the DIW process and by casting the ink formulations in silicone molds, undergoing the drying and curing procedures described in Subsection 2.7. Five samples were tested to ensure statistical significance. Tensile properties were measured using a Zwick Roell Z010 machine (ZwickRoell GmbH & Co. KG, Ulm, Germany) with a 10 kN load cell, operating at a speed of 10 mm/min.

After the tensile test, the fracture surface morphology of the PVA25P and PVA25C samples was analyzed using SEM. The micrographs were obtained with a Cambridge Stereoscan 360 system (Cambridge Instrument Company Ltd., Cambridge, UK). The sample surfaces were prepared by applying a physical vapor deposition of gold for 1 min. Three fracture surfaces were analyzed for each sample type.

3. Results and discussion

3.1. Leather filler

Microscope images of non-sieved and sieved particles are reported in Fig. 4 a-c and d-f, respectively. The particles display an irregular morphology, consisting of fibers of various sizes, i.e., in the order of hundreds of micrometers to millimeters. This fact is consistent with the shredding process used to produce them. The sieving process does not significantly modify the morphology of the particles, as they continue to appear as fiber entanglements even after sieving.

Regarding the Weibull distribution of particle dimensions and shape factor, the results are presented in Fig. 4g and Fig. 4h, with the corresponding Weibull coefficients listed in Table 3. According to the results, the sieving process reduces the average particle size and decreases data dispersion. In addition, the sieving process alters the shape factor, resulting in an increase towards a value closer to 1. This value suggests that the sieved particles tend to be more spherical and less elongated compared to the non-sieved particles, reducing the variability of the filler in terms of morphology.

The thermal properties of LF have been analyzed using DSC and TGA (Fig. 5a). The DSC analysis showed that the leather filler remains stable up to 120 °C. At 120 °C and 220 °C, two endothermic peaks were observed, likely corresponding to collagen denaturation. A prominent degradation peak appeared at 310 °C. TGA analysis confirmed that the LF particles exhibit thermal stability up to 200 °C, with significant mass loss (~70 %) between 200 °C and 450 °C, indicating the onset and progression of thermal degradation. The primary degradation peak was noted at 320 °C, closely matching the DSC results.

ATR-FTIR spectrum of the filler is reported in Fig. 5b. Leather is composed of collagen proteins and lipids, which contain NH and OH functional groups [75]. FTIR analysis reveals characteristic peaks corresponding to the Amide A, B, I, II, and III bands. The Amide A band, located at 3310 cm^{-1}, is primarily associated with N—H stretching vibrations, while the Amide *B* band at 3077 cm^{-1} typically arises from N—H vibrations in Fermi resonance with the Amide II band. The Amide *I* band, observed at 1637 cm^{-1}, predominantly corresponds to the stretching vibrations of C═O, whereas the Amide II band, located at 1548 cm^{-1}, is attributed to N—H bending and C—N stretching vibrations. The Amide III band, appearing at 1236 cm^{-1}, is more complex and involves C═N stretching coupled with N—H bending and C—H stretching [76].

A peak near 1660 cm^{-1}, typically associated with C═N bonds formed via Schiff base reactions during glutaraldehyde tanning, was not observed in the FTIR spectra. This absence is likely due to an overlap with the Amide I band. The presence of OH groups in leather was expected; a broad band between 3000 and 3500 cm^{-1} overlaps with Amide A and Amide *B*, making their signals difficult to distinguish [76]. The presence of numerous polar groups in leather, as observed in the FTIR spectra, enhances its compatibility with the PVA matrix, which is rich in OH groups. The abundance of polar groups in both materials promotes strong intermolecular interactions, such as hydrogen bonding, ensuring

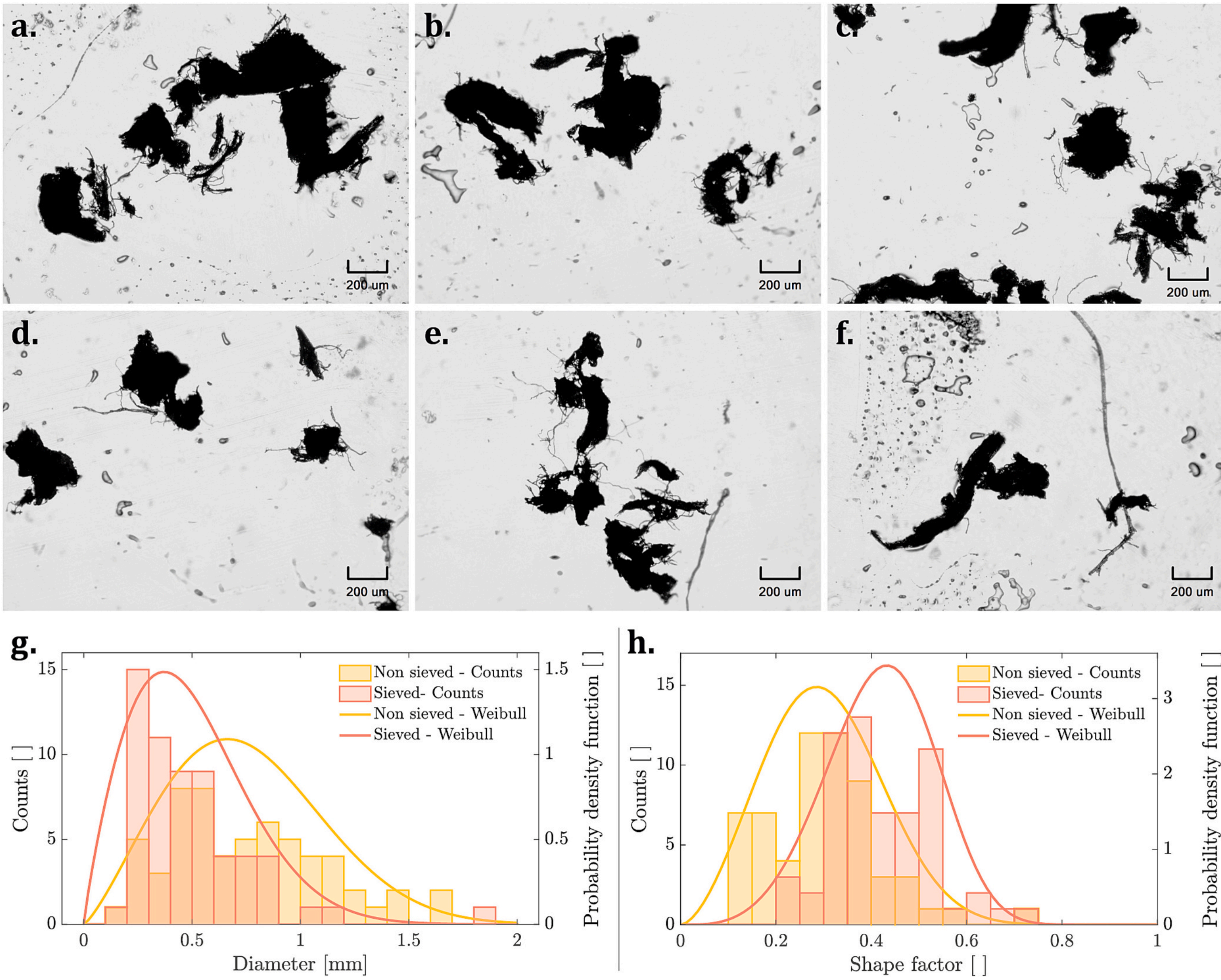

Fig. 4. LF particle characterization: (a), (b), (c), micrographs of the non-sieved LF particles at 50× magnification; (d), (e), (f) micrographs of the sieved LF particles (LF-S) at 50× magnification; (g) distribution graphs of the particle diameters and (h) of the shape factor according to the Weibull distribution.

Table 3
Coefficients of LF particle distribution of diameters and shape factors according to the Weibull distribution function: scale factor λ and shape factor k for sieved and non-sieved LF.

Distribution	LF sieving	Scale factor λ	Shape factor k
Particle diameter	Sieved particles (LF-S)	0.86	2.26
	Non-sieved particles (LF)	0.55	1.87
Particle SF	Sieved particles (LF-S)	0.34	2.69
	Non-sieved particles (LF)	0.46	4.18

effective integration. This compatibility not only facilitates uniform blending but also contributes to good performance in terms of mechanical behavior, as the cohesive interactions between the leather-derived components and the PVA matrix improve the overall structural integrity of the material.

3.2. Matrix synthesis

The degree of methacrylation is defined as the percentage of -OH groups on the PVA chain that underwent reaction with GMA, bonding one methacrylic group. The ^{1}H NMR spectra of PVA dissolved in DMSO-d_6 show the three separate signals at 4.6, 4.4, and 4.2 ppm, which are assigned to the proton in the (-OH) group of PVA [77].

Protons belonging to the methyl group in the methacrylic pendants show a peak at 1.86 ppm. In addition, two multiplet signals in the ranges 2.629–2.638 ppm and 2.759–2.777 ppm correspond to the two protons of the -CH_2 group in the epoxy ring [70]. The presence of such peaks would correspond to the presence of residual GMA or reaction by-products.

The degree of substitution can be calculated by integrating the aforementioned peaks and comparing their intensities. Existing three protons in each methyl group, the intensity of the corresponding peak has to be divided by 3.

$$DM[\%] = \frac{\frac{I_{1.86}}{3}}{I_{OH} + \frac{I_{1.86}}{3}} \cdot 100 \tag{6}$$

Fig. 6 shows the ^{1}H NMR spectra of PVA and PVA. The high-intensity signal of the solvent ($(CD_3)_2SO$) in both spectra is located at 3.3 ppm. In the PVA-M spectrum, the peak relative to protons belonging to the methyl group in the methacrylic pendants is visible at 1.9 ppm. This peak is not present in the PVA spectrum, confirming methacrylation. In the range between 4.2 and 4.4 ppm, three peaks corresponding to protons in –OH groups are visible in both spectra.

The DM of the synthesized PVA-M was calculated to quantify the reaction's effectiveness, resulting equal to 4.46 %. In order to perform

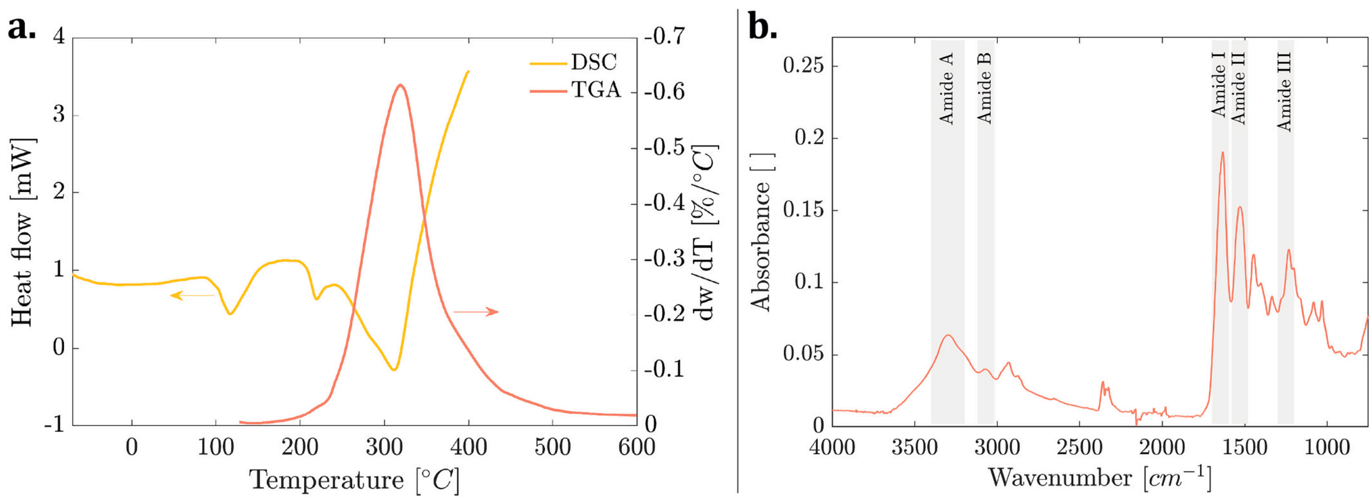

Fig. 5. (a) Thermal analyses performed on leather filler (DSC and TGA); (b) ATR-FTIR spectrum of the leather filler.

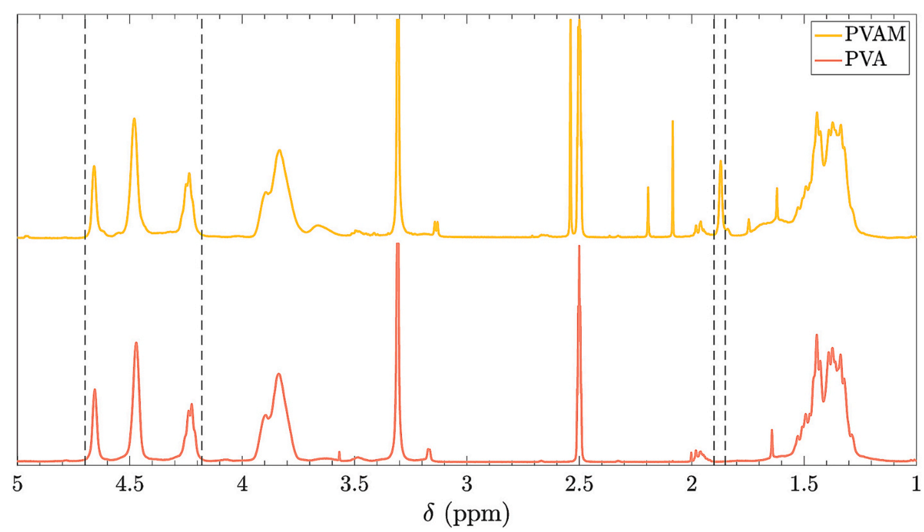

Fig. 6. ^{1}H NMR spectra of PVA-M (top) and PVA matrixes (bottom).

the methacrylation reaction, glycidyl methacrylate has been added in a quantity equal to 5 % mol/mol OH. Therefore, the maximum DM achievable is 5 %. Having a DM equal to 4.46 % indicates a reaction yield equal to 89 %. Finally, the absence of peaks in the ranges 2.629–2.638 ppm and 2.759–2.777 ppm indicates that no residual GMA or reaction by-products are present in the polymer matrix. Therefore, acetone washings following the methacrylation properly removed any contaminant.

The reaction yield confirms that the desired methacrylic groups have been successfully added to the polymer, rendering it suitable for cross-linking and subsequent processing. The lack of reaction residues further ensures safe handling.

The self-polymerization of GMA via epoxy ring opening is a possible side reaction. However, previous literature examples do not account for this; thus, ^{1}H NMR was assumed to be a reliable method for determining the degree of methacrylation of PVA [70,71,78,79].

3.3. Rheological behavior characterization

The rheological behavior of a material is a crucial parameter for its suitability in DIW [47]. In detail, shear-thinning behavior is highly desirable as it facilitates flow through the nozzle. The presence of a yield point is also essential to ensure that the material stops flowing once deposited onto the build plate. Rapid return to a solid-like state after large deformation is also necessary to prevent the material from continuing to flow after deposition, as well as to ensure shape retention after extrusion.

Fig. 7 provides a detailed overview of the rheological characterization results for the LF-PVA ink formulations. The complex viscosity was measured as a function of oscillation frequency using frequency sweep tests. According to Fig. 7c, all formulations exhibit shear-thinning behavior, where the viscosity decreases with increasing frequency (and therefore shear rate). This behavior is particularly important for DIW applications, as it facilitates smoother extrusion through the nozzle by reducing resistance during the 3D printing process. Furthermore, the curves show an increase in the complex viscosity when increasing the percentage content of LF-S particles of the inks, achieving comparable slopes in the three formulations. The consistent slopes observed in the frequency sweep curves indicate the same fundamental mechanisms governing the rheological behavior of the three formulations (Fig. 7c). This uniformity suggests that the shear-thinning properties arise from similar interactions within each formulation. The primary difference in the viscosity can be due to the variation in solid content, with higher solid content leading to higher viscosity.

The storage and loss moduli as a function of shear deformation were evaluated through amplitude sweep tests (Fig. 7a). In these tests, the

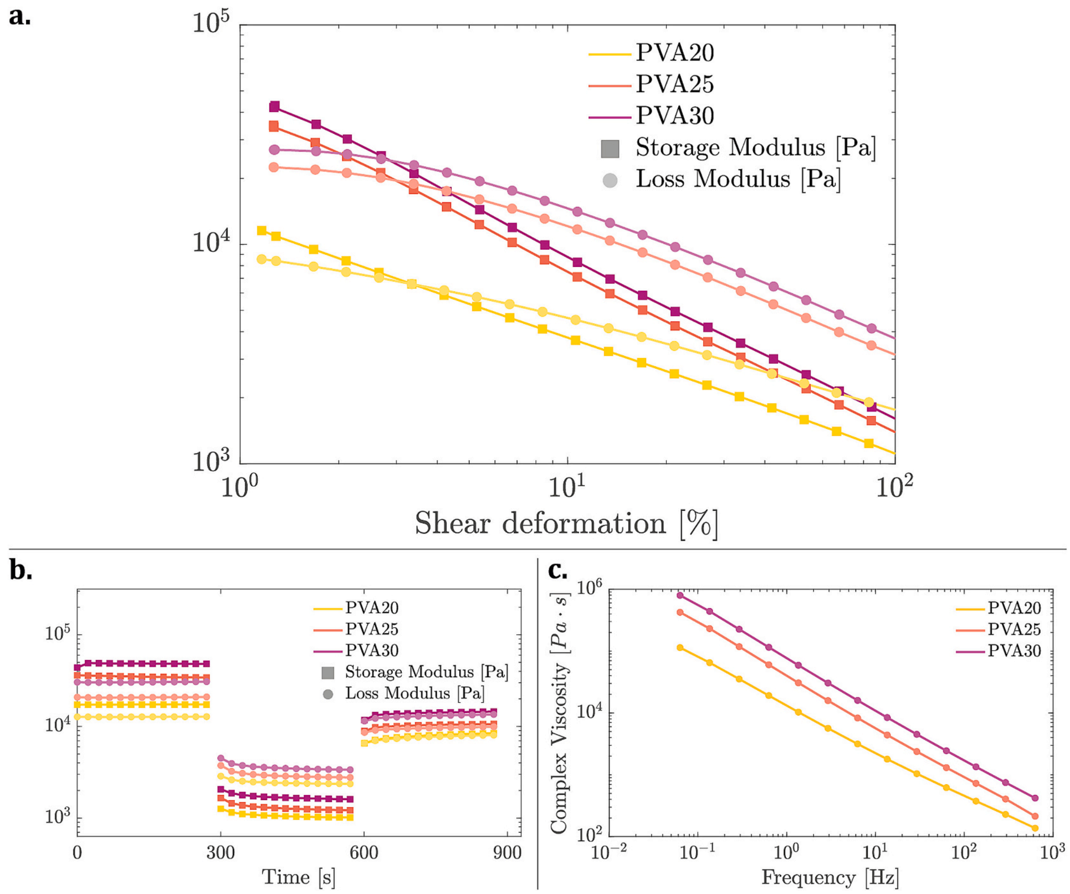

Fig. 7. Results of rheological test performed on PVA composites: (a) Storage and loss moduli as a function of the shear deformation obtained from the amplitude sweep test: (b) Storage and loss moduli as a function of time obtained from the three-intervals test: (c) Complex viscosity as a function of frequency obtained from the frequency test.

storage modulus (G') and loss modulus (G") provide insights into the material's mechanical behavior under deformation. A higher storage modulus compared to the loss modulus indicates a solid-like behavior, which is essential in DIW to prevent unwanted flow after extrusion and ensure shape retention. Conversely, when the loss modulus exceeds the storage modulus, the material exhibits a fluid-like behavior, which is fundamental for a consistent extrusion through the nozzle. [47] The point where the storage and loss moduli intersect is known as the yield point, representing the transition from solid-like to fluid-like behavior. As visible in Fig. 7a, the three formulations demonstrated a distinct yield point, with the yield deformation decreasing as the content of LF-S increased. This trend is expected, as higher LF content makes the ink more paste-like rather than liquid, enhancing its solid-like properties.

The oscillatory stress at the yield point is referred to as yield stress, a critical parameter in determining the material's ability to maintain its shape under stress. The formulations with 20 % wt., 25 % wt., and 30 % wt. LF content exhibited yield stresses of 312 Pa, 785 Pa, and 949 Pa, respectively. These relatively high yield stress values suggest that the inks can ensure good shape fidelity after the extrusion, making them well-suited for DIW applications. The increase in yield stress with higher LF-S content further confirms that the inks become more rigid and resistant to deformation, enhancing their expected performance in additive manufacturing processes, especially in terms of shape retention. Furthermore, the increase in LF-S content not only raises the yield stress but also leads to a corresponding increase in the storage and loss moduli.

Finally, the storage and loss moduli were measured across three distinct intervals to simulate the extrusion process. The first interval involved applying a 1 % shear deformation, the second a 100 % deformation, and the third a return to 1 % deformation. This experimental setup is designed to mimic the different stages of DIW, specifically the quasi-static residence of the ink within the cartridge, the high deformation experienced during extrusion through the nozzle, and the static resting of the ink on the build plate after the extrusion, respectively. For successful shape retention during DIW, the ink should quickly recover its solid-like behavior after the high deformation phase experienced during the extrusion. This rapid recovery ensures that the 3D printed material maintains its structural integrity and does not deform or flow undesirably after extrusion. Fig. 7b presents the results of this test for the three different ink formulations. All formulations exhibit a solid-like behavior in the first interval, as indicated by the storage modulus being higher than the loss modulus. This behavior is essential during the quasi-static residence phase to prevent premature flow of the ink from the cartridge. In the second interval, the formulations shift to a viscous regime, characterized by a higher loss modulus necessary for smooth extrusion through the nozzle. In the third interval, following the large deformation, all three formulations immediately revert to a solid-like state, with the storage modulus again surpassing the loss modulus. This rapid return to a solid-like behavior is highly promising for DIW, as it can be linked with inks' excellent shape fidelity and ensure shape retention when 3D printing parts.

3.4. 3D Printing

Fig. 8a and Fig. 8b show the two geometries utilized to evaluate the shape fidelity and printability of the inks, the 3 × 3 grid, and the shape for filament fusion test, in this case, using PVA20. As shown in the figures, both the grid pattern and adjacent filaments were successfully deposited, maintaining their intended geometries compared to the

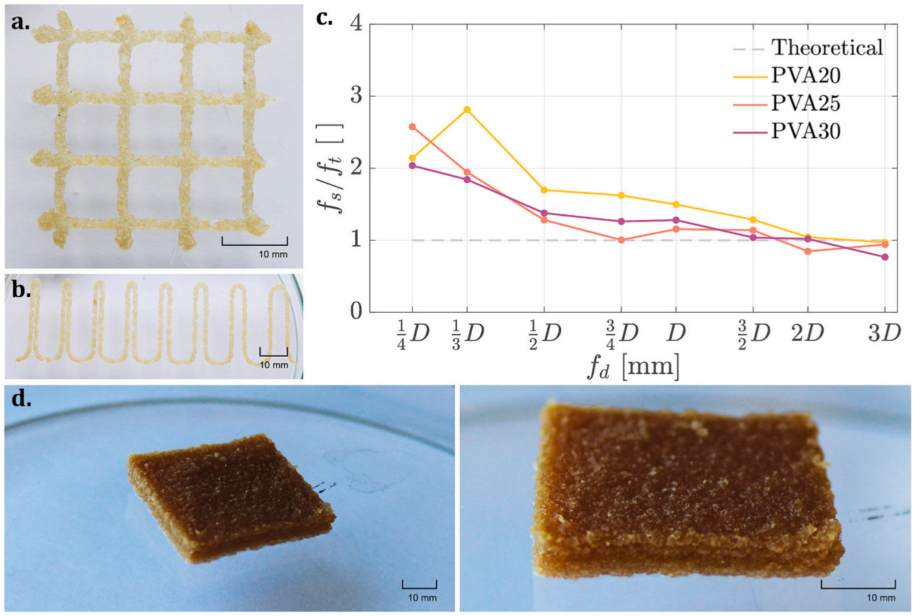

Fig. 8. Printability evaluation of the PVA-Leather inks for DIW: shape fidelity test of the (a) 3 × 3 grid obtained with the PVA20 formulation; (b) filament fusion test geometry obtained with the PVA20 formulation; (c) Measured points from the filament fusion test; and (d) a 3D printed squared multilayered sample made with the PVA25 formulation.

toolpaths (Fig. 3). The ink achieved a consistent extrusion during its deposition, indicating a smooth and uninterrupted flow during the 3D printing process.

Regarding the 3 × 3 grid structure, the Pr (pore retention) values were found to be 1.06, 1.12, and 1.24 for PVA20, PVA25, and PVA30, respectively. These values indicate good printability of the formulations, as the pores retained their intended shapes without becoming excessively rounded after the deposition. However, Pr values greater than 1 suggest that the nozzle may drag some material when it passes over previously extruded toolpaths. This effect appears to be more pronounced with higher LF content, as evidenced by the increasing Pr values. Consequently, the greater the LF content in the formulation, the more material is likely to be dragged by the nozzle during the 3D printing process, potentially leading to slight shape modifications in the final geometry. This phenomenon is consistent with the rheological properties of the inks (Subsection 3.3). As the modulus increases, the ink behaves more like a solid, making it more prone to being pulled along by the nozzle when it crosses over the previously deposited paths. Nevertheless, the overall 3D printing fidelity remains high, with the printed structures maintaining their intended forms.

In the filament fusion test, all three formulations successfully produced distinct filaments even at the smallest spacing (1/4D, Fig. 3c), as indicated by the presence of all data points on the graph in Fig. 8c. This result demonstrates excellent shape fidelity across the formulations, with the inks effectively maintaining separation between filaments despite the toolpath proximity. The ability to achieve such precision, even at minimal toolpath spacing, is crucial for producing detailed high-resolution 3D printed structures. When examining the ratio of filament spacing to filament thickness (f_s/f_t), PVA20 shows slightly inferior performance compared to PVA25 and PVA30, which maintain a ratio closer to 1 across almost all f_d values (filament distance). A ratio closer to 1 indicates the extrusion shape retention in the loop region, suggesting better control over the deposition process. The slightly lower performance of PVA20 may be attributed to its rheological properties, as lower viscosity and modulus can lead to a less controlled deposition and a slower return to the solid-like state. Nevertheless, the differences between the formulations are minimal, achieving good printability and shape retention of the extruded paths.

The PVA25 was then used to 3D print a squared multilayered sample as a proof-of-concept to demonstrate the feasibility of achieving higher dimensions in the z-axis direction. As visible in Fig. 8d, the part was successfully 3D printed, confirming the suitability of the ink formulations to be used for multilayered 3D-shaped parts.

The printability assessment also confirmed that the use of water-based formulations allowed the processing of materials containing a high scrap fraction without any issue related to the 3D printing process, maximizing the use of recycled materials and reducing the need for virgin resources.

3.5. *Curing characterization*

The crosslinking yield obtained from DSC measurements was 72 %, 91 %, and 89 % for PVA20, PVA25, and PVA30, respectively. According to the values, the crosslinking reaction does not achieve complete yield, which is in line with the characteristics of the polyvinyl alcohol (PVA) used. Specifically, the PVA in these formulations has a low degree of methacrylation, meaning that there are relatively few reactive methacrylate groups available for crosslinking. These reactive groups are also attached to long polymer chains, further complicating the crosslinking process of the PVA matrix. The limited number of reactive sites, coupled with their attachment to high molecular weight chains, creates a situation where the reactive groups may not easily encounter one another, thereby hindering the crosslinking reaction. The high molecular weight of the PVA chains restricts the mobility of the reactive groups, reducing their tendency to likelihood that they will come into contact and form crosslinks. This inherent limitation in the system means that even with extended curing times or increased amounts of curing agents, the crosslinking yield may not significantly improve.

Nevertheless, a crosslinking yield below 100 % does not indicate a failure in the crosslinking process. Despite the incomplete reaction, the system can still achieve a high gel content and degree of crosslinking. The high gel content corresponds to a network structure that is effectively crosslinked, with minimal presence of unreacted or free polymer chains. Accordingly, the crosslinking process, although not entirely complete, is sufficient to produce a stable material with the desired mechanical and chemical properties. The gel content of the formulations was determined using the formula reported in Subsection 2.8, yielding results of 99 %, 99 %, and 98 % for PVA20, PVA25, and PVA30, respectively. These high gel content values indicate the effectiveness of the curing process, ensuring the crosslinking of most of the polymer network of the matrixes.

The crosslinking process in these formulations does not begin with low molecular weight monomers but with high molecular weight PVA. As a result, even if a small percentage of non-crosslinked polymer is still present, it does not significantly impact the mechanical properties of the final material. In contrast to formulations derived from low molecular weight monomers, where incomplete crosslinking might lead to significant weaknesses or safety concerns, the nature of high molecular weight PVA means that the presence of non-crosslinked polymer is less critical. Additionally, the non-crosslinked material does not pose health risks upon contact, as it is simply PVA, which is a widely used, non-toxic material. Although the mechanical properties and safety are not significantly compromised by the presence of a small amount of non-crosslinked PVA, achieving near-complete crosslinking is still important to prevent the material from being soluble in water since incomplete crosslinking could lead to dissolution, reducing the durability of the material.

The two analyses revealed that nearly all polymer molecules are effectively incorporated into the network, as evidenced by the high gel content measurements, although not all methacrylic groups can react during the curing process. This result indicates that the curing process has successfully formed a robust polymer network despite the incomplete reaction of methacrylic groups. Consequently, the curing process can be considered successful and adopted for the UV-assisted DIW process.

3.6. Mechanical properties

Fig. 9 and Fig. 10, and Table 4 present the results of the mechanical testing, highlighting the performance of the different compositions. All tested formulations exhibit a higher modulus and tensile strength compared to the control composition, which means PVA0S. This fact suggests a homogeneous filler dispersion within the polymer matrix, demonstrating good adhesion compatibility and effective impregnation

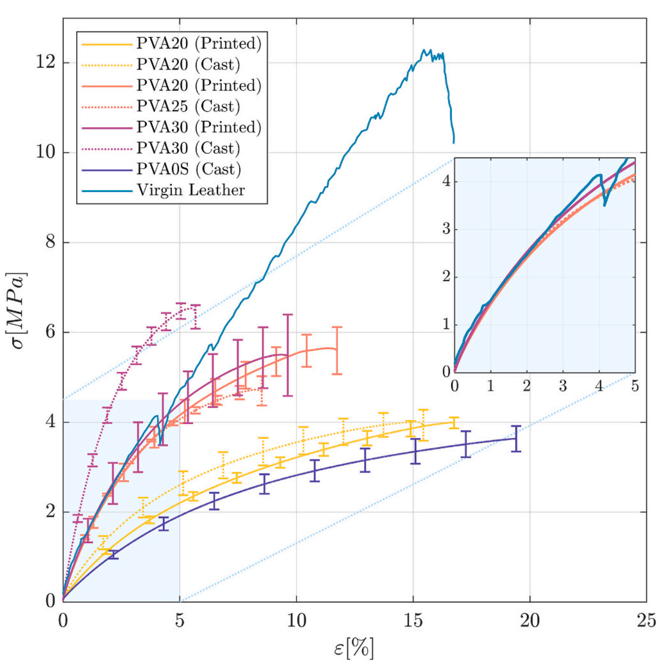

Fig. 9. Mean stress-strain curves from the different batches of the PVA-leather tensile specimens obtained by casting and DIW (PVA0S, PVA20, PVA 25, PVA30) compared with experimental data of virgin leather, reproduced from Li et al., 2009 [6]. The nested plot provides a more detailed view of the initial deformation region.

even without functionalization. The filler thus acts as a reinforcing agent, enhancing the mechanical properties of the material.

The 3D printed specimens exhibit different behavior, showing increased elongation at break alongside a lower elastic modulus. This result indicates the influence of the 3D printing process on the material properties, which can be due to a preferential particle alignment during the extrusion through the nozzle in DIW processes. Given the branched morphology of leather particles, this alignment could lead to an apparent reduction in the effective volumetric fraction of the filler, thereby diminishing its reinforcing effect. Nevertheless, the alignment along the axial direction likely contributes to the increased elongation at break, enhancing the toughness of the 3D printed material compared to conventional manufacturing processes.

The SEM images of the fracture surfaces of the samples provide further insight into the structural differences between the 3D printed and cast formulations (Fig. 11). The morphology of the filler qualitatively matches the findings from the initial characterization of the raw LF samples (Subsection 3.1). The filler dimensions in the SEM analysis are consistent with those previously measured, further confirming the preservation of the filler's characteristics throughout the processing. According to Fig. 11, the filler is homogeneously dispersed within the matrix, which supports the reinforcing effect observed in the tensile tests. The images show no significant voids or signs of segregation, further confirming the uniform distribution of the filler throughout the material.

At higher magnification (Fig. 11d), the 3D printed sample reveals a preferential alignment of the LF particles, particularly in the direction normal to the fracture surface. This alignment is consistent with the mechanical history experienced by the ink during the 3D printing process, where the extrusion through a nozzle likely orients the filler particles along the flow direction. Indeed, such an effect is not visible in the cast sample (Fig. 11b). Previous works in the literature have already reported the enhancing mechanism of filler particles due to extrusion-induced orientation in DIW processes [51,67].

According to the SEM images, the apparent volumetric fraction of the filler in the 3D printed samples is lower compared to the cast samples. This observation suggests that during the 3D printing process, the alignment of filler particles results in a less branched configuration of the LF-S. This alignment leads to an apparent reduction in the volumetric fraction within the matrix, which may explain the observed differences in mechanical properties. Specifically, this alignment contributes to the increased elongation at break and overall toughness of the 3D printed material, as the aligned fillers can better absorb and dissipate stress along the axial direction. In contrast, the cast sample shows a markedly different microstructure. The LF-S particles in the cast sample appear more disordered and have a disordered "fluffy" appearance, indicating no significant alignment during the casting process. This disordered arrangement allows for a higher apparent volumetric fraction of the filler. These findings underscore the significant impact of the selected UV-DIW 3D printing process on the microstructure and, consequently, on the mechanical behavior of these composite materials.

The objective of these formulations was to replicate the mechanical characteristics of virgin leather, whose mechanical behavior is reported in Fig. 9 [6]. It shows a Young's modulus of approximately 100 MPa, a stress at break of 12 MPa, and a strain at break of 15 %, but first damage occurs at 4 % strain. Among the tested formulations, PVA25P, PVA25C, and PVA30P most closely align with these criteria. Indeed, their mechanical behavior is identical to that of virgin leather up to 4 % strain, while exhibiting greater elastic deformation at strains exceeding 4 %, where leather has already experienced damage. In particular, PVA25P has a deformation at break as high as 12 %, showing the highest toughness among tested formulations. By experimenting with different combinations of plasticizers and fillers, these target properties can be overcome, further optimizing the material to match the desired mechanical performance of virgin leather and allowing the formulations to withstand higher stresses and strains before failure.

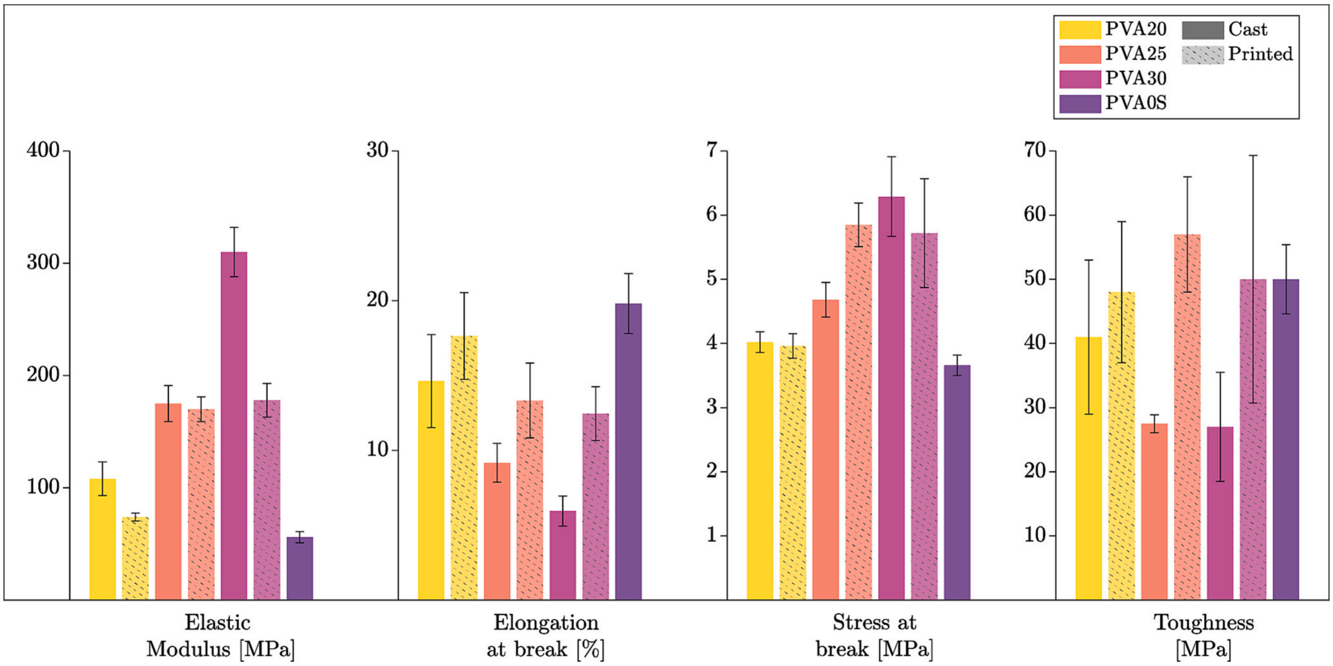

Fig. 10. Experimental values from the tensile tests on the different PVA-leather formulations: (a) elastic modulus, (b) elongation at break; (c) stress at break; and (d) toughness.

Table 4
Experimental values and standard deviations from the tensile tests on the different PVA-leather formulations.

Sample	Elastic modulus [MPa]	Elongation at break [%]	Stress at break [MPa]	Toughness [MPa]
PVA20C	108.3 ± 15.5	14.6 ± 3.1	4.0 ± 0.2	41.4 ± 11.8
PVA20P	74.1 ± 3.5	17.6 ± 2.9	3.9 ± 0.2	48.0 ± 11.1
PVA25C	175.4 ± 16.5	9.2 ± 1.3	4.8 ± 0.3	27.5 ± 1.4
PVA25P	170.8 ± 11.0	13.3 ± 2.6	5.9 ± 0.9	57.5 ± 19.3
PVA30C	310.0 ± 22.6	6.0 ± 1.1	6.3 ± 0.6	26.8 ± 8.5
PVA30P	178.6 ± 15.0	12.5 ± 1.8	5.7 ± 0.9	50.1 ± 19.4
PVA0SC	56.2 ± 5.1	19.8 ± 2.1	3.7 ± 0.2	50.3 ± 5.4

4. Conclusions

This paper focused on the characterization and printability of photo- and thermal-curable PVA-based inks filled with LF scraps from the tanning industry for UV-assisted DIW. The study aims to recycle impactful raw materials by producing composites with mechanical properties comparable to virgin leather, offering the potential to reduce reliance on primary resources and deliver environmental benefits. The selected technology, UV-assisted direct ink writing, is particularly suitable due to its association with reduced energy consumption, particle-preferred alignment, and scrap content optimization, as it operates as a cold extrusion process.

The morphology and thermal properties of the LF scraps were assessed through optical microscopy and DSC. A PVA-based matrix was synthesized for three ink formulations with different LF content, reaching up to ~20 % w/w_{matrix} and almost 40 % w/w_{tot}. The rheological behavior of the different ink formulations was studied to assess their printability, which was then quantified through 3D printed samples. The curing process and the crosslinking degree of the inks were measured through DSC and gel content measurements to validate the use of cross-linkable LF-based inks for UV-DIW. Tensile tests were finally carried out to define the mechanical properties of the formulations, understanding the influence of the 3D printing process on the final parts and the reinforcing nature of the filler through SEM micrographs on the fracture surfaces.

Despite the preferential elongated shape of LF, the sieved LF particles (LF-S) can be successfully used as fillers for novel ink formulations for DIW. As from the ^{1}H NMR analysis, the methacrylation made the PVA-based matrix suitable for cross-linkable ink formulations and, hence, for UV-DIW processes. This fact was consistent with the curing characterization of the ink formulations, reaching crosslinking yields of ~90 % through DSC and gel content measures up to 99 %, indicating the presence of a robust polymer network after the 3D printing and curing processes. According to the rheological tests, the PVA-LF formulations show a shear-thinning behavior with increasing complex viscosities at higher LF contents. They exhibit a distinct transition from solid-like to fluid-like behavior when shear deformations are applied, e.g., during extrusion, as well as rapid recovery of the solid-like behavior to ensure shape retention, making them suitable for DIW processes. The printability tests later confirmed the good shape retention of the inks, even at minimal toolpath spacing. The best results were obtained with the PVA25P formulation, also used to 3D print a multilayered sample as a proof-of-concept of 3D parts. Tensile tests and SEM analysis also demonstrated the reinforcing effect of the filler, as well as its homogeneous dispersion in the matrix, achieving a preferential alignment of the LF particles in the 3D printed samples thanks to the DIW extrusion process. The micrographs confirmed the overall quality of the UV-DIW process and the preferred alignment, with no distinguishable extruded paths and a limited amount of voids in the fracture surface cross-sections. Among the tested inks, the PVA25P, PVA25C, and PVA 30P formulations demonstrated mechanical properties most similar to those of virgin leather, especially until 4 % strain, widening the potential applications of these inks.

Leather scraps can be a viable opportunity to implement new circular value chains in the leather industry, reducing the use of virgin materials and paving the way for new tailored applications, and maximizing leather scrap use through AM processes, in this case, UV-DIW. This work presents a novel approach to considering leather shavings as secondary raw materials, reusing abundant shredded scraps for new materials with properties close to virgin leather. Future works should deepen the characterization of these inks, e.g., performing additional mechanical tests or recyclability at their end-of-life. Moreover, studying new matrixes can increase the range of potential applications, e.g., developing

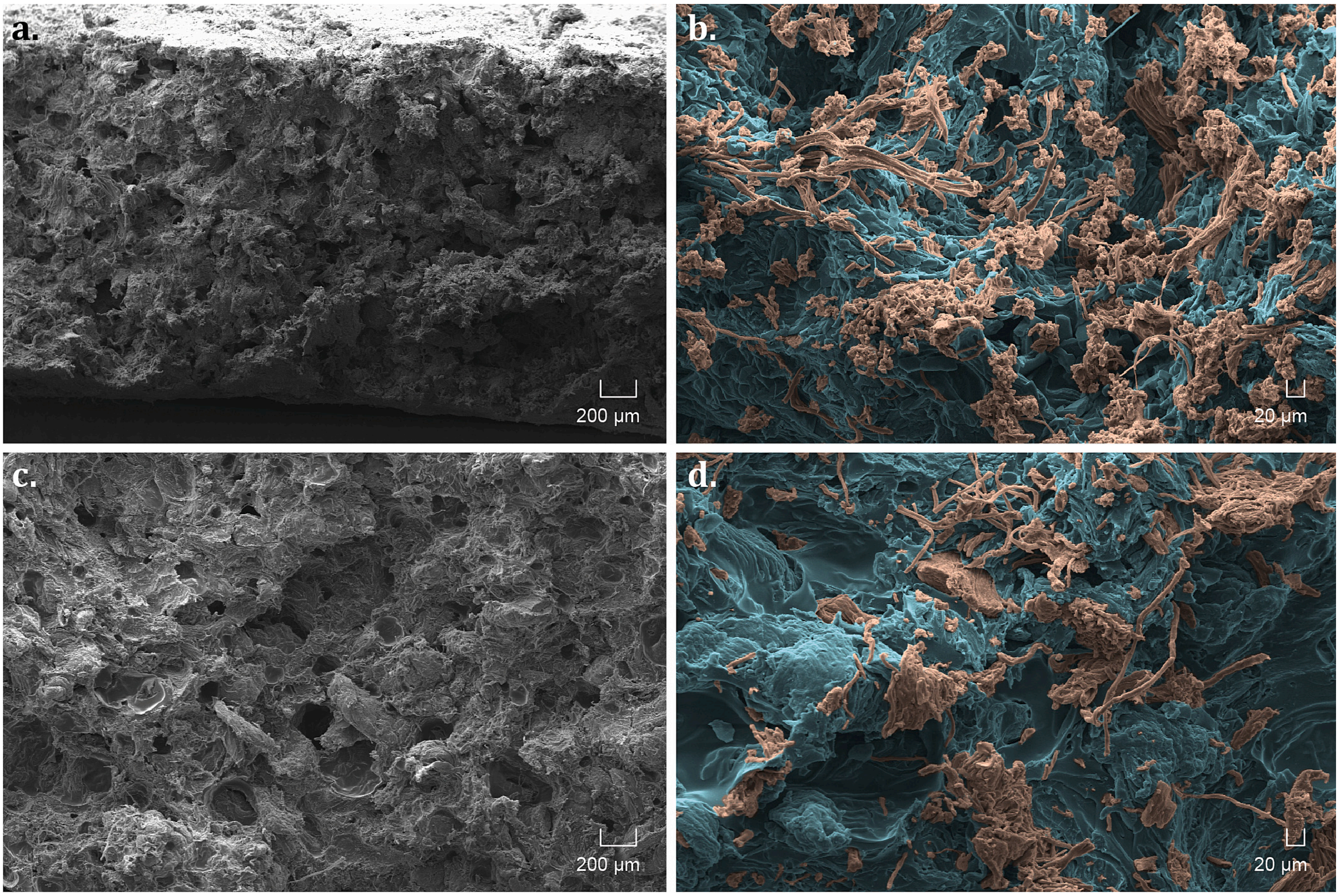

Fig. 11. SEM micrographs of the fracture surfaces from the tensile samples of the PVA-leather formulations: (a) obtained by casting, (b) highlighting the PVA matrix (in blue) and LF-S particles (in orange); (c) obtained by DIW, (d) highlighting the PVA matrix (in blue) and LF-S particles (in orange). Matrix and filler particles have been identified through Energy Dispersive Spectroscopy (EDS), and colors have been added accordingly.

flexible LF-based materials or getting closer results to virgin leather by changing the ink formulation. Finally, other leather scraps can be considered for future experimentation, e.g., from different processing steps, and an in-depth study on the achievable geometries should be carried out to increase the Technology Readiness Level of this solution. Despite the current limitations in the achievable shape complexity and dimensions, these inks can increase the application sectors for DIW and leather scraps, e.g., detecting new market niches or generating closed-loop solutions within the leather industry value chain, reducing the need for virgin raw materials and the environmental impact of this industrial sector.

CRediT authorship contribution statement

Luca Guida: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alessia Romani:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Davide Negri:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Marco Cavallaro:** Writing – review & editing, Supervision. **Marinella Levi:** Writing – review & editing, Validation, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used OpenAI ChatGPT 4.0 to improve the readability and language of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to acknowledge the Italian Leather Research Institute (SSIP, Pozzuoli, Naples, Italy) for kindly providing the milled glutaraldehyde-tanned leather shavings used for the experimental work. This study was carried out within the MICS (Made in Italy Circular and Sustainable) Extended Partnership and received funding from the European Union Next GenerationEU (Piano Nazionale di Ripresa e Resilienza (PNRR) Missione 4 Componente 2, Investimento 1.3 D.D. 1551.11

10 2022, PE00000004). This manuscript reflects only the authors' views and opinions, neither the European Union nor the European Commission can be considered responsible for them.

References

[1] L. Muthukrishnan, Nanotechnology for cleaner leather production: a review, Environ. Chem. Lett. 19 (2021) 2527–2549, https://doi.org/10.1007/s10311-020-01172-w.
[2] S. Lu, Y. Feng, P. Zhang, W. Hong, Y. Chen, H. Fan, D. Yu, X. Chen, Preparation of flame-retardant polyurethane and its applications in the leather industry, Polymers (Basel) 13 (2021) 1730, https://doi.org/10.3390/polym13111730.
[3] E.K. Roh, K.W. Oh, S.H. Kim, Classification of synthetic polyurethane leather by mechanical properties according to consumers' preference for fashion items, Fibers Polymers 14 (2013) 1731–1738, https://doi.org/10.1007/s12221-013-1731-x.
[4] A.G. Ward, The mechanical properties of leather, Rheol. Acta 13 (1974) 103–112, https://doi.org/10.1007/BF01526892.
[5] C.-K. Liu, N.P. Latona, G.L. DiMaio, P.H. Cooke, Viscoelasticity studies for a fibrous collagen material: chrome-free leather, J. Mater. Sci. 42 (2007) 8509–8516, https://doi.org/10.1007/s10853-007-1744-1.
[6] Z. Li, D. Paudecerf, J. Yang, Mechanical behaviour of natural cow leather in tension, Acta Mechanica Solida Sinica 22 (2009) 37–44, https://doi.org/10.1016/S0894-9166(09)60088-4.
[7] Leather Goods Market Size, Share & COVID-19 Impact Analysis, By Source (Full Grain Leather and Synthetic Leather), By Product (Apparel, Luggage, Footwear, and Others), By End-user (Men, Women, and Kids), and Regional Forecast, 2023–2030, Source: https://www.fortunebusinessinsights.com/leather-goods-market-104405, 2024.
[8] K. Chojnacka, D. Skrzypczak, K. Mikula, A. Witek-Krowiak, G. Izydorczyk, K. Kuligowski, P. Bandrów, M. Kułażyński, Progress in sustainable technologies of leather wastes valorization as solutions for the circular economy, J. Clean. Prod. 313 (2021) 127902, https://doi.org/10.1016/j.jclepro.2021.127902.
[9] J. Hu, Z. Xiao, R. Zhou, W. Deng, M. Wang, S. Ma, Ecological utilization of leather tannery waste with circular economy model, J. Clean. Prod. 19 (2011) 221–228, https://doi.org/10.1016/j.jclepro.2010.09.018.
[10] V. Muralidharan, S. Palanivel, M. Balaraman, Turning problem into possibility: a comprehensive review on leather solid waste intra-valorization attempts for leather processing, J. Clean. Prod. 367 (2022) 133021, https://doi.org/10.1016/j.jclepro.2022.133021.
[11] K. Chojnacka, D. Skrzypczak, K. Mikula, A. Witek-Krowiak, G. Izydorczyk, K. Kuligowski, P. Bandrów, M. Kułażyński, Progress in sustainable technologies of leather wastes valorization as solutions for the circular economy, J. Clean. Prod. 313 (2021) 127902, https://doi.org/10.1016/j.jclepro.2021.127902.
[12] S. Famielec, Chromium concentrate recovery from solid tannery waste in a thermal process, Materials 13 (2020) 1533, https://doi.org/10.3390/ma13071533.
[13] D. Navarro, J. Wu, W. Lin, P. Fullana-i-Palmer, R. Puig, Life cycle assessment and leather production, J. Leather Sci. Eng. 2 (2020) 26, https://doi.org/10.1186/s42825-020-00035-y.
[14] D. Masilamani, B. Madhan, G. Shanmugam, S. Palanivel, B. Narayan, Extraction of collagen from raw trimming wastes of tannery: a waste to wealth approach, J. Clean. Prod. 113 (2016) 338–344, https://doi.org/10.1016/j.jclepro.2015.11.087.
[15] L. Šánek, J. Pecha, K. Kolomazník, M. Bařinová, Biodiesel production from tannery fleshings: feedstock pretreatment and process modeling, Fuel 148 (2015) 16–24, https://doi.org/10.1016/j.fuel.2015.01.084.
[16] S. Majee, G. Halder, T. Mandal, Formulating nitrogen-phosphorous-potassium enriched organic manure from solid waste: a novel approach of waste valorization, Process Saf. Environ. Prot. 132 (2019) 160–168, https://doi.org/10.1016/j.psep.2019.10.013.
[17] H. Chen, Y. Li, H. Dai, L. Chen, X. Ding, Z. Hu, Collagen hydrolyzed extract derived from leather waste as a multifunctional additive for the preparation of granular fertilizer, Sustain. Chem. Pharm. 36 (2023) 101327, https://doi.org/10.1016/j.scp.2023.101327.
[18] M. Ayele, D.Y. Limeneh, T. Tesfaye, W. Mengie, A. Abuhay, A. Haile, G. Gebino, A review on utilization routes of the leather industry biomass, Adv. Mater. Sci. Eng. 2021 (2021), https://doi.org/10.1155/2021/1503524.
[19] M. Parisi, A. Nanni, M. Colonna, Recycling of chrome-tanned leather and its utilization as polymeric materials and in polymer-based composites: a review, Polymers (Basel) 13 (2021) 429, https://doi.org/10.3390/polym13030429.
[20] S. Selvaraj, S. Ramalingam, S. Parida, J.R. Rao, N.F. Nishter, Chromium containing leather trimmings valorization: sustainable sound absorber from collagen hydrolysate intercalated electrospun nanofibers, J. Hazard. Mater. 405 (2021) 124231, https://doi.org/10.1016/j.jhazmat.2020.124231.
[21] M.E. Bautista, L. Pérez, M.T. García, S. Cuadros, A. Marsal, Valorization of tannery wastes: Lipoamino acid surfactant mixtures from the protein fraction of process wastewater, Chem. Eng. J. 262 (2015) 399–408, https://doi.org/10.1016/j.cej.2014.10.004.
[22] D. Reike, W.J.V. Vermeulen, S. Witjes, The circular economy: new or refurbished as CE 3.0? — exploring controversies in the conceptualization of the circular economy through a focus on history and resource value retention options, Resour. Conserv. Recycl. 135 (2018) 246–264, https://doi.org/10.1016/j.resconrec.2017.08.027.
[23] O. Awogbemi, D.V. Von Kallon, K.A. Bello, Resource recycling with the aim of achieving zero-waste manufacturing, Sustainability 14 (2022) 4503, https://doi.org/10.3390/su14084503.
[24] Md. Tauhiduzzaman, Md.A. Mottalib, M.J. Rahman, Md.A. Kalam, Preparation and characterization of composite sheets from solid leather waste with plant fibers: a waste utilization effort, Clean Technol. Environ. Policy 26 (2024) 1025–1038, https://doi.org/10.1007/s10098-023-02642-9.
[25] A. Battig, G. Sanchez-Olivares, D. Rockel, M. Maldonado-Santoyo, B. Schartel, Waste not, want not: the use of leather waste in flame retarded EVA, Mater. Des. 210 (2021) 110100, https://doi.org/10.1016/j.matdes.2021.110100.
[26] E. Kiliç, Q. Tarrés, M. Delgado-Aguilar, X. Espinach, P. Fullana-i-Palmer, R. Puig, Leather waste to enhance mechanical performance of high-density polyethylene, Polymers (Basel) 12 (2020) 2016, https://doi.org/10.3390/polym12092016.
[27] M.M. Owen, E.O. Achukwu, I.O. Arukalam, M. Muhammad, A.Z. Romli, Effect of processing temperatures on the thermal and mechanical properties of leather waste-ABS composites, Compos. Adv. Mater. 30 (2021), https://doi.org/10.1177/26349833211060056.
[28] B. Ramaraj, Mechanical and thermal properties of ABS and leather waste composites, J. Appl. Polym. Sci. 101 (2006) 3062–3066, https://doi.org/10.1002/app.24113.
[29] A. Laurentia, D. Viorica, S. Maria, G. Mihai, N. Mihaela, D.D. Corina, P. Xioyan, Biodegradable polymer composite based on recycled polyurethane and finished leather waste, IOP Conf Ser Earth Environ. Sci. 401 (2019) 012006, https://doi.org/10.1088/1755-1315/401/1/012006.
[30] E.T. Musa, Hamza Abba, A.S. Ahmed, Investigation of the mechanical and morphological properties of high-density polyethylene (Hdpe)/leather waste composites, IOSR, J. Appl. Chem. 10 (2017) 48–58, https://doi.org/10.9790/5736-1001014858.
[31] E. Kılıç, P. Fullana-i-Palmer, M. Fullana, M. Delgado-Aguilar, R. Puig, Circularity of new composites from recycled high density polyethylene and leather waste for automotive bumpers, Test. Perform. Environ. Impact Sci. Total Environ. 919 (2024) 170413, https://doi.org/10.1016/j.scitotenv.2024.170413.
[32] E. Kiliç, H. Oliver-Ortega, Q. Tarrés, M. Delgado-Aguilar, P. Fullana-i-Palmer, R. Puig, Valorization strategy for leather waste as filler for high-density polyethylene composites: analysis of the thermal stability, insulation properties and chromium leaching, Polymers (Basel) 13 (2021) 3313, https://doi.org/10.3390/polym13193313.
[33] L. Sartore, F. Bignotti, S. Pandini, A. D'Amore, L. Di Landro, Green composites and blends from leather industry waste, Polym. Compos. 37 (2016) 3416–3422, https://doi.org/10.1002/pc.23541.
[34] S. Sharma, P. Sudhakara, J. Singh, M.R. Sanjay, S. Siengchin, Fabrication of novel polymer composites from leather waste fibers and recycled poly(ethylene-vinyl-acetate) for value-added products, Sustainability 15 (2023) 4333, https://doi.org/10.3390/su15054333.
[35] A. Battig, G. Sanchez-Olivares, D. Rockel, M. Maldonado-Santoyo, B. Schartel, Waste not, want not: the use of leather waste in flame retarded EVA, Mater. Des. 210 (2021) 110100, https://doi.org/10.1016/j.matdes.2021.110100.
[36] E.T. Musa, Hamza Abba, A.S. Ahmed, Investigation of the mechanical and morphological properties of high-density polyethylene (Hdpe)/leather waste composites, IOSR, J. Appl. Chem. 10 (2017) 48–58, https://doi.org/10.9790/5736-1001014858.
[37] C. Abeykoon, A. McMillan, B.K. Nguyen, Energy efficiency in extrusion-related polymer processing: a review of state of the art and potential efficiency improvements, Renew. Sustain. Energy Rev. 147 (2021) 111219, https://doi.org/10.1016/j.rser.2021.111219.
[38] A. Romani, V. Rognoli, M. Levi, Design, materials, and extrusion-based additive manufacturing in circular economy contexts: from waste to new products, Sustainability 13 (2021) 7269, https://doi.org/10.3390/su13137269.
[39] H.A. Colorado, E.I.G. Velásquez, S.N. Monteiro, Sustainability of additive manufacturing: the circular economy of materials and environmental perspectives, J. Mater. Res. Technol. 9 (2020) 8221–8234, https://doi.org/10.1016/j.jmrt.2020.04.062.
[40] A. Bhatia, A.K. Sehgal, Additive manufacturing materials, methods and applications: a review, Mater. Today Proc. 81 (2023) 1060–1067, https://doi.org/10.1016/j.matpr.2021.04.379.
[41] ASTM International, Standard Terminology for Additive Manufacturing – General Principles – Terminology, 2022.
[42] F.A. Cruz Sanchez, H. Boudaoud, M. Camargo, J.M. Pearce, Plastic recycling in additive manufacturing: a systematic literature review and opportunities for the circular economy, J. Clean. Prod. 264 (2020) 121602, https://doi.org/10.1016/j.jclepro.2020.121602.
[43] A. Yousaf, A. Al Rashid, R. Polat, M. Koç, Potential and challenges of recycled polymer plastics and natural waste materials for additive manufacturing, sustainable, Mater. Technol. 41 (2024) e01103, https://doi.org/10.1016/j.susmat.2024.e01103.
[44] A. Romani, R. Suriano, M. Levi, Biomass waste materials through extrusion-based additive manufacturing: a systematic literature review, J. Clean. Prod. 386 (2023) 135779, https://doi.org/10.1016/j.jclepro.2022.135779.
[45] M. Ateeq, M. Shafique, A. Azam, M. Rafiq, A review of 3D printing of the recycled carbon fiber reinforced polymer composites: processing, potential, and perspectives, J. Mater. Res. Technol. 26 (2023) 2291–2309, https://doi.org/10.1016/j.jmrt.2023.07.171.
[46] E. Manarin, O. Boumezgane, A. Giannino, V. De Fabritiis, G. Griffini, S. Turri, Towards a zero-waste chemcycling of thermoset polymer composites: catalyst assisted mild solvolysis for clean carbon fiber liberation and circular coating

development, Sustain. Mater. Technol. 41 (2024) e01031, https://doi.org/10.1016/j.susmat.2024.e01031.
[47] A. Mantelli, A. Romani, R. Suriano, M. Levi, S. Turri, Direct ink writing of recycled composites with complex shapes: process parameters and ink optimization, Adv. Eng. Mater. 23 (2021), https://doi.org/10.1002/adem.202100116.
[48] M. Sauerwein, E.L. Doubrovski, Local and recyclable materials for additive manufacturing: 3D printing with mussel shells, Mater. Today Commun. 15 (2018) 214–217, https://doi.org/10.1016/j.mtcomm.2018.02.028.
[49] B. Coppola, C. Tardivat, S. Richaud, J.-M. Tulliani, L. Montanaro, P. Palmero, 3D printing of dense and porous alkali-activated refractory wastes via direct ink writing (DIW), J. Eur. Ceram. Soc. 41 (2021) 3798–3808, https://doi.org/10.1016/j.jeurceramsoc.2021.01.019.
[50] A. Romani, S. Caba, R. Suriano, M. Levi, Recycling glass and carbon fibers for reusable components in the automotive sector through additive manufacturing, Appl. Sci. 13 (2023) 5848, https://doi.org/10.3390/app13105848.
[51] K. Niendorf, B. Raeymaekers, Additive manufacturing of polymer matrix composite materials with aligned or organized filler material: a review, Adv. Eng. Mater. 23 (2021), https://doi.org/10.1002/adem.202001002.
[52] S. Bagatella, L. Guida, G. Scagnetti, E. Gariboldi, M. Salina, N. Galimberti, L. Castoldi, M. Cavallaro, R. Suriano, M. Levi, Tailoring thermal conductivity and printability in boron nitride/epoxy nano- and micro-composites for material extrusion 3D printing, Polymer 317 (2025) 127899, https://doi.org/10.1016/j.polymer.2024.127899.
[53] A. Asghar, K. Khan, M.S. Rashid, M. Hamza, Z. Liu, C. Liu, Z. Chen, Extrusion-based additive manufacturing of carbonaceous and non-carbonaceous electrode materials for electrochemical energy storage devices, Adv. Mater. Technol. (2024), https://doi.org/10.1002/admt.202400136.
[54] E. Gokce Bahcegul, E. Bahcegul, N. Ozkan, 3D printing of crude lignocellulosic biomass extracts containing hemicellulose and lignin, Ind. Crops Prod. 186 (2022) 115234, https://doi.org/10.1016/j.indcrop.2022.115234.
[55] E.G. Bahçegül, E. Bahçegül, N. Özkan, 3D printing of Hemicellulosic biopolymers extracted from lignocellulosic agricultural wastes, ACS Appl. Polym. Mater. 2 (2020) 2622–2632, https://doi.org/10.1021/acsapm.0c00256.
[56] A. Alhelal, Z. Mohammed, S. Jeelani, V.K. Rangari, 3D printing of spent coffee ground derived biochar reinforced epoxy composites, J. Compos. Mater. 55 (2021) 3651–3660, https://doi.org/10.1177/00219983211002237.
[57] K. Pitt, O. Lopez-Botello, A.D. Lafferty, I. Todd, K. Mumtaz, Investigation into the material properties of wooden composite structures with in-situ fibre reinforcement using additive manufacturing, Compos. Sci. Technol. 138 (2017) 32–39, https://doi.org/10.1016/j.compscitech.2016.11.008.
[58] A. Sarmah, S.K. Desai, A.G. Crowley, G.C. Zolton, G.B. Tezel, E.M. Harkin, T. Q. Tran, K. Arole, M.J. Green, Additive manufacturing of nanotube-loaded thermosets via direct ink writing and radio-frequency heating and curing, Carbon 200 (2022) 307–316, https://doi.org/10.1016/j.carbon.2022.08.063.
[59] R. Hamaied, C. Gao, A. Spagnoli, F. Berto, C. Bertolin, Wrinkling of soft bilayers created with additive manufacturing: experimental tests, finite element modelling and analytical validation, Procedia Struct. Integrity 47 (2023) 102–112, https://doi.org/10.1016/j.prostr.2023.06.046.
[60] L. Mogas-Soldevila, G. Matzeu, M. Lo Presti, F.G. Omenetto, Additively manufactured leather-like silk protein materials, Mater. Des. 203 (2021) 109631, https://doi.org/10.1016/j.matdes.2021.109631.
[61] I.K.M. Yu, K.-H. Wong, Food waste-derived 3D printable materials: a carbon neutral solution to global foodloss, Trends Food Sci. Technol. 137 (2023) 156–166, https://doi.org/10.1016/j.tifs.2023.05.014.
[62] A. Baroutaji, A. Arjunan, J. Robinson, M. Ramadan, M.A. Abdelkareem, A. Vance, A. Arafat, A.-G. Olabi, Developments and prospects of additive manufacturing for thermoelectric materials and technologies, Sustain. Mater. Technol. 41 (2024) e01008, https://doi.org/10.1016/j.susmat.2024.e01008.
[63] M. Monisha, M. Anand, S. Panigrahi, M.V.W. Kofoed, R. Aghababaei, S. Agarwala, Bioinspired poly(vinyl alcohol) films with tunable adhesion and self-healing for biodegradable electronics and beyond, Sustain. Mater. Technol. 41 (2024) e01084, https://doi.org/10.1016/j.susmat.2024.e01084.
[64] M. Kaur, S. Sharma, A. Kalia, S.K. Jawandha, Maize starch-PVA nanocomposite biodegradable antimicrobial packaging films for enhancement of shelf-life of Agaricus bisporus, Sustain. Mater. Technol. 41 (2024) e01114, https://doi.org/10.1016/j.susmat.2024.e01114.
[65] J.W. Kemp, A.A. Diaz, E.C. Malek, B.P. Croom, Z.D. Apostolov, S.R. Kalidindi, B. G. Compton, L.M. Rueschhoff, Direct ink writing of ZrB2-SiC chopped fiber ceramic composites, Addit. Manuf. 44 (2021) 102049, https://doi.org/10.1016/j.addma.2021.102049.
[66] M. Zhang, Y. Wang, M. Jian, C. Wang, X. Liang, J. Niu, Y. Zhang, Spontaneous alignment of graphene oxide in hydrogel during 3D printing for multistimuli-responsive actuation, Adv. Sci. 7 (2020), https://doi.org/10.1002/advs.201903048.
[67] G. Siqueira, D. Kokkinis, R. Libanori, M.K. Hausmann, A.S. Gladman, A. Neels, P. Tingaut, T. Zimmermann, J.A. Lewis, A.R. Studart, Cellulose nanocrystal inks for 3D printing of textured cellular architectures, Adv. Funct. Mater. 27 (2017), https://doi.org/10.1002/adfm.201604619.
[68] C.A. Schneider, W.S. Rasband, K.W. Eliceiri, NIH image to ImageJ: 25 years of image analysis, Nat. Methods 9 (2012) 671–675, https://doi.org/10.1038/nmeth.2089.
[69] T.M. Zobeck, T.W. Popham, E.L. Skidmore, J.A. Lamb, S.D. Merrill, M.J. Lindstrom, D.L. Mokma, R.E. Yoder, Aggregate-mean diameter and wind-erodible soil predictions using dry aggregate-Size distributions, Soil Sci. Soc. Am. J. 67 (2003) 425–436, https://doi.org/10.2136/sssaj2003.4250.
[70] A.K. Elbadawy, M.O. Ahmed, N.K. Sherine, M.A. Heba, A.E. Ali, In-situ UV-photopolymerized PVA-g-GMA hydrogels for biomedical applications: I. Synthesis, characterizations and grafting optimization, J. Appl. Pharm. Sci. (2018), https://doi.org/10.7324/JAPS.2018.8106.
[71] E.A. Kamoun, A.M. Omer, M.M. Abu-Serie, S.N. Khattab, H.M. Ahmed, A. A. Elbardan, Photopolymerized PVA-g-GMA hydrogels for biomedical applications: factors affecting hydrogel formation and bioevaluation tests, Arab. J. Sci. Eng. 43 (2018) 3565–3575, https://doi.org/10.1007/s13369-017-3054-5.
[72] A. Mantelli, A. Romani, R. Suriano, M. Diani, M. Colledani, E. Sarlin, S. Turri, M. Levi, UV-assisted 3D printing of polymer composites from thermally and mechanically recycled carbon fibers, Polymers 13 (2021) 726, https://doi.org/10.3390/polym13050726.
[73] A. Schwab, R. Levato, M. D'Este, S. Piluso, D. Eglin, J. Malda, Printability and shape Fidelity of bioinks in 3D bioprinting, Chem. Rev. 120 (2020) 11028–11055, https://doi.org/10.1021/acs.chemrev.0c00084.
[74] ASTM International, Standard Test Method for Tensile Properties of Polymer Matrix Composite Materials, 2014.
[75] K.H. Sizeland, R.L. Edmonds, M.M. Basil-Jones, N. Kirby, A. Hawley, S. Mudie, R. G. Haverkamp, Changes to collagen structure during leather processing, J. Agric. Food Chem. 63 (2015) 2499–2505, https://doi.org/10.1021/jf506357j.
[76] Y. Hu, J. Liu, L. Luo, X. Li, F. Wang, K. Tang, Kinetics and mechanism of thermal degradation of aldehyde tanned leather, Thermochim. Acta 691 (2020) 178717, https://doi.org/10.1016/j.tca.2020.178717.
[77] I. Korbag, S. Mohamed Saleh, Studies on the formation of intermolecular interactions and structural characterization of polyvinyl alcohol/lignin film, Int. J. Environ. Stud. 73 (2016) 226–235, https://doi.org/10.1080/00207233.2016.1143700.
[78] J. Sinna, R. Jeencham, P. Mueangkhot, S. Sophon, P. Noralak, R. Raksapakdee, P. Numpaisal, Y. Ruksakulpiwat, Development of poly(vinyl alcohol) grafted Glycidyl methacrylate/cellulose nanofiber injectable hydrogels for Meniscus tissue engineering, Polymers 15 (2023) 4230, https://doi.org/10.3390/polym15214230.
[79] A.A. Morgacheva, A.A. Artyukhov, A.V. Panov, M.G. Gordienko, M.I. Shtil'man, Ya O. Mezhuev, Synthesis of polyvinyl alcohol with methacrylate groups and of hydrogels based on it, Russ. J. Appl. Chem. 88 (2015) 617–621, https://doi.org/10.1134/S1070427215040102.